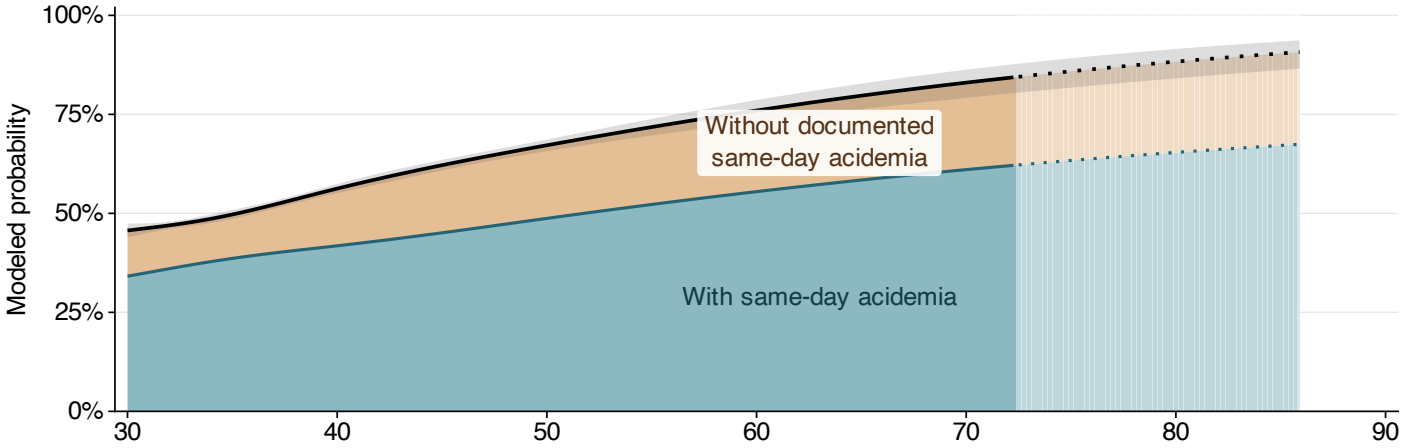
BMI and hypercapnia with and without same-day acidemia

Same recorded-day co-occurrence; not verified same-sample pairing

Shared model/reference N = 22,195; with acidemia: 8,809; without: 2,860

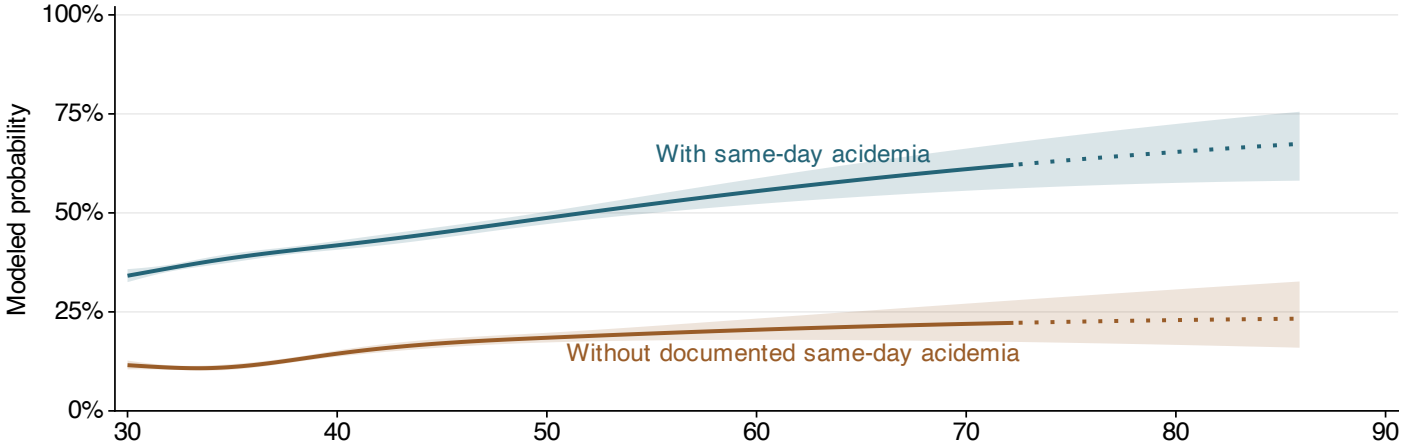
A Total hypercapnia, divided into its two components

Band thickness = component probability; top boundary = total (95% CI)

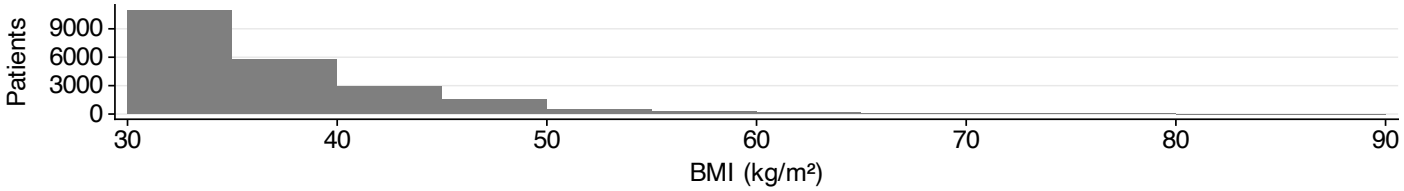


B Component curves on a common baseline

Pointwise 95% intervals; dotted segments mark sparse local evidence



C Patients in the shared co-observed population



Supplemental Figure S1. Same recorded-day co-occurrence; not verified same-sample pairing. Primary fixed-index patients with both arterial analytes share inherited adjustment and the primary BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

Joint clinical supplement; investigator interpretation and S1 approval pending.